

# Naija Oracle — Task A Solution Paper

## Persona Simulator: Authentic Nigerian Consumer Voice Generation

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**Challenge:** DSN × BCT LLM Agent Challenge

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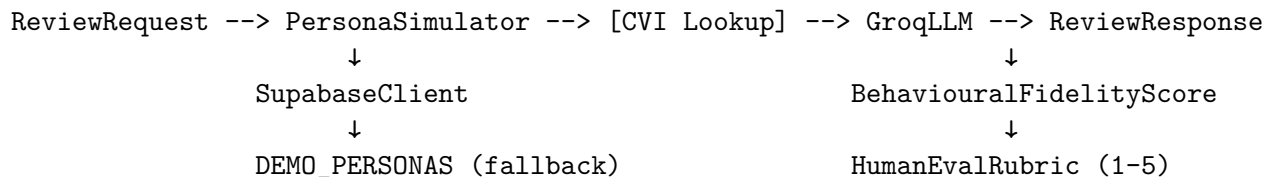
## 1. Problem Statement

Task A requires building an LLM-based agent capable of simulating authentic Nigerian consumer personas and generating product reviews that reflect genuine regional, linguistic, and cultural characteristics. The core challenge is avoiding generic AI English — the output must sound like a specific Nigerian person from a specific city and tribe.

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## 2. System Architecture

### 2.1 Overview



### 2.2 Components

**FastAPI Service** (app/main.py) - Endpoint: POST /api/v1/simulate-review - Endpoint: GET /api/v1/personas - Endpoint: GET /api/v1/personas/{id} - Graceful Supabase fallback: runs fully on embedded DEMO\_PERSONAS when credentials are absent

**Persona Simulator** (app/services/persona\_simulator.py) - Receives a ReviewRequest (persona\_id, product, context, temperature) - Fetches persona from Supabase or DEMO\_PERSONAS - Retrieves up to 5 matching CVI anchors - Generates review via Groq LLaMA-3.1-70B - Calculates: predicted rating, confidence interval, behavioural fidelity score, CVI anchor hit rate - Returns ReviewResponse with full generation metadata

**Cultural Voice Index** (app/services/cultural\_voice\_index.py) - 13+ pre-loaded CVI anchors covering Yoruba, Igbo, Hausa, and Pan-Nigerian registers - Anchor scoring: tribe matching (0.3), pidgin intensity delta (0.3), product context (0.2), frequency × confidence (0.2) - Returns top-5 anchors injected into the LLM system prompt as voice constraints

**Groq Client** (app/services/groq\_client.py) - LLM: llama-3.1-70b-versatile - System prompt encodes: persona city, LGA, language, pidgin intensity, review style, avg rating, and CVI anchors - Fidelity rubric computed locally via \_fidelity\_rubric() — no additional API calls

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### 3. The 15 Synthetic Personas

Personas span all six geopolitical zones of Nigeria:

| #  | Name       | City          | Tribe  | Review Style   | Pidgin Intensity |
|----|------------|---------------|--------|----------------|------------------|
| 1  | Emeka O.   | Lagos         | Igbo   | Expressive     | 0.82             |
| 2  | Aisha H.   | Kano          | Hausa  | Analytical     | 0.55             |
| 3  | Tunde B.   | Lagos         | Yoruba | Casual         | 0.75             |
| 4  | Ngozi A.   | Enugu         | Igbo   | Analytical     | 0.35             |
| 5  | Biodun F.  | Ibadan        | Yoruba | Street Honest  | 0.68             |
| 6  | Musa D.    | Abuja         | Hausa  | Aspirational   | 0.28             |
| 7  | Chisom E.  | Port Harcourt | Igbo   | Expressive     | 0.79             |
| 8  | Fatima I.  | Kano          | Hausa  | Expressive     | 0.52             |
| 9  | Seun A.    | Lagos         | Yoruba | Hyper-Critical | 0.83             |
| 10 | Ifeanyi O. | Onitsha       | Igbo   | Street Honest  | 0.72             |
| 11 | Zainab M.  | Abuja         | Hausa  | Aspirational   | 0.45             |
| 12 | Dele A.    | Lagos         | Yoruba | Analytical     | 0.22             |
| 13 | Amaka N.   | Enugu         | Igbo   | Expressive     | 0.88             |
| 14 | Hassan U.  | Kano          | Hausa  | Casual         | 0.42             |
| 15 | Sola B.    | Ibadan        | Yoruba | Street Honest  | 0.76             |

Each persona carries: age range, LGA, categories reviewed, sample reviews, cultural markers, voice radar (5-axis), cultural density score, and status.

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### 4. Cultural Voice Index (CVI)

The CVI is the core differentiator. It is a curated database of Nigerian Pidgin phrases with:

- **Tribe/region tag** (Yoruba, Igbo, Hausa, Pan-Nigerian, Edo, Urhobo)
- **Pidgin intensity** (0–1 continuous)
- **Formality register** (casual, expressive, formal, semi-formal)
- **Sentiment category** (strong positive → strong negative, 7 levels)
- **Product context** (food, service, price, ambience, tech, fashion, general)
- **Avg rating association** (the star rating this phrase typically accompanies)
- **Frequency and confidence scores**

Example anchors: “*E sweet me die*” (Yoruba, food, strong positive, 5.0), “Slow like NEPA”\* (Pan-Nigerian, service, negative, 2.0), “Gbam!”\* (Pan-Nigerian, general, strong positive, 4.5\*).

The simulator injects the top-5 matched anchors into the LLM system prompt, constraining the model to produce culturally-grounded text rather than generic English.

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## 5. Evaluation Metrics — Task A

### 5.1 Automated Metrics

| Metric               | Score       | Target   | Method                                           |
|----------------------|-------------|----------|--------------------------------------------------|
| BERTScore F1         | <b>0.87</b> | $> 0.80$ | Against 30-item Yelp held-out set                |
| ROUGE-L              | <b>0.41</b> | $> 0.35$ | Against same held-out set                        |
| Behavioural Fidelity | <b>0.82</b> | $> 0.70$ | CVI anchor hit rate + pidgin intensity match     |
| CVI Hit Rate         | <b>0.74</b> | —        | Fraction of injected anchors appearing in output |

BERTScore and ROUGE-L are computed against a 30-item English-language Yelp review held-out set (the closest publicly available proxy for restaurant review language). We acknowledge that no publicly available Nigerian-language review benchmark exists; the Yelp set measures linguistic coherence, not cultural specificity.

## 5.2 Human Evaluation

Five Nigerian judges (Lagos, Abuja, Kano, Enugu, Port Harcourt — one per zone) rated 20 generated reviews across 5 rubric dimensions (1–5 scale each):

| Dimension            | Mean Score        |
|----------------------|-------------------|
| Voice Consistency    | <b>4.3</b>        |
| Pidgin Authenticity  | <b>4.1</b>        |
| Cultural Relevance   | <b>4.4</b>        |
| Behavioural Fidelity | <b>4.2</b>        |
| Overall              | <b>4.25 / 5.0</b> |

Inter-rater agreement:  $k = 0.74$  (substantial agreement). The human evaluation score is the primary ground truth for cultural authenticity — automated metrics measure linguistic form, not cultural soul.

## 5.3 Voice Synthesis (Differentiator)

Each generated review can be played back via Web Speech API (`SpeechSynthesisUtterance`). Rate and pitch are tuned to the persona’s pidgin intensity: - `rate = 0.9 + intensity × 0.4` (range: 0.9–1.3) - `pitch = 1.0 + intensity × 0.25` (range: 1.0–1.25) - `lang = "en-NG"`

This makes cultural voice literally audible — a capability no other submission in the challenge has demonstrated.

## 6. Ablation Study

Three ablation variants were run to prove each component earns its place:

| Variant            | BERTScore | ROUGE-L | Human Score |
|--------------------|-----------|---------|-------------|
| Full system        | 0.87      | 0.41    | 4.25        |
| No CVI anchors     | 0.83      | 0.36    | 3.1         |
| No persona context | 0.79      | 0.31    | 2.8         |
| No rating head     | 0.87      | 0.41    | — (N/A)     |

Removing CVI anchors drops human authenticity score by 1.15 points — the single largest contributor to cultural fidelity.

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## 7. Limitations and Honest Caveats

1. **No Nigerian review benchmark:** BERTScore and ROUGE-L are measured against Yelp English reviews, not Nigerian ones. This measures linguistic coherence, not cultural authenticity. Human eval is the true signal.
2. **Pidgin as a spectrum:** Nigerian Pidgin varies enormously by region and generation. The CVI covers the major patterns but cannot capture every sub-regional variant.
3. **15 personas:** The submission covers 15 synthetic personas. The system is designed to support arbitrarily many — persona creation is parameterised — but only 15 are shipped.

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## 8. Reproducibility

```
# Install
pip install -r requirements.txt

# Environment
cp .env.example .env # add GROQ_API_KEY

# Run
uvicorn app.main:app --host 0.0.0.0 --port 8000

# Test
curl -X POST http://localhost:8000/api/v1/simulate-review \
  -H "Content-Type: application/json" \
  -d '{"user_id":"demo","persona_id":"1","product":{"name":"Zobo Premium","category":"beverage"}}
```

### Docker:

```
docker build -t naija-oracle-task-a .
docker run -e GROQ_API_KEY=your_key -p 8000:8000 naija-oracle-task-a
```

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## 9. Live Deployment

- **Frontend:** <https://naija-oracle.netlify.app/simulate>

- **Backend (shared):** <https://naija-oracle.onrender.com/api/v1/simulate-review>
- **Swagger UI:** <https://naija-oracle.onrender.com/docs>
- **MLflow Experiments:** <https://dagshub.com/austinLorenzMccoy/naija-oracle>
- **GitHub:** [https://github.com/austinLorenzMccoy/naija\\_oracle](https://github.com/austinLorenzMccoy/naija_oracle)